

backend/core/token_budget.py Documentation (Roman Urdu)

Yeh file application ka "Hisaab Kitaab (Accountant)" k tor par kaam krti hai jo Tokens (AI words/lafz) process hone ki limit ginanti ha. Agar proper calculation ni hogy tou AI apny jawab ki middle m pause ho giga y limit se bahir chalker error laiga.

File Ka Maqsad (Purpose)

Bade NLP Models (LLMs) k liay Tokens gin-nay waly software like tiktoken bohot bhari (heavy) hotay hein. Yeh file aik lite-weight logic or clever asool (heuristics) use karti hai tokens budget map or estimate karny key liya without loading huge memory requirements.

Code Blocks aur Functions ki Tafseel

1. Regex List (`_SUMMARY_PATTERNS`, `_DETAIL_PATTERNS`)

- File ke upper wale hsse me user queries ko map karny wali dictionaries haen. Jese "Summarize", "List", "Compare", ta k query ko category dia jye.

2. `estimate_token_count(text)`

- **Tafseel:** Text string leta ha aur rough size return (English text k mutbiq takriban har 4 character py 1 token lgts ha, jis mn 30% addition word count sa fix hota ha, overall average speed barray calculation algorithm sy 10 guna zyada hy usky).

3. `classify_query(query) / estimate_required_tokens()`

- **Logic:** Agar user ne sirf "yes/no/define" bola hai, to category short hoti ha aur output budget 200 token limit rakh de jati ha. Agar user nay "explain in depth step by step" likhaa hai to `very_long` category pr lagakar limits ko extend (2500) krdia jata ha!

4. `class TokenTracker`

- **Kya karta hai?:** Realtime streaming ma hisaab track karti hai k ab tk kitne Token generated hain aur bacha huway max quota kitna reh gha ha (utilization block output).